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ACADEMY

Video 1 · Number Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 1–6 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

6	19	Q1–Q6	19:50
questions	marks	E1.1 – E1.18	video runtime

M1 method mark A1 accuracy mark (needs the M mark) B1 independent mark

Question 1

Prime factors and square numbers

(a) Write 756 as a product of its prime factors. [2]

Keep dividing by the smallest prime that works:

2	756
2	378
3	189
3	63
3	21
7	7
	1

A correct division ladder (or factor tree) with at least two correct primes **M1**

$756 = 2^2 \times 3^3 \times 7$ **A1**

Watch out: 756 = 4 × 189 is not the answer, because 4 is not prime. Also accepted: 2 × 2 × 3 × 3 × 3 × 7. A list without × signs may lose the A mark.

(b) Smallest positive integer k so that $756k$ is a square number. [1]

In a square number, **every power is even**.

- 2^2 is already even ✓
- 3^3 needs one more 3, making 3^4
- 7^1 needs one more 7, making 7^2

$k = 3 \times 7$

$k = 21$ **B1**

Check: $756 \times 21 = 15\,876 = 126^2$ ✓

Question 2

Recurring decimals and fractions

(a) Write 0.2$\dot{1}$ as a fraction. [2]

The dot is only on the 1, so the number is 0.2111…

Let x = 0.2111…

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 10x =  2.1111...
100x = 21.1111...
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Subtract so the recurring part cancels: **M1**

$100x - 10x = 21.111\ldots - 2.111\ldots$

$90x = 19$

$x = \frac{19}{90}$ **A1**

Show the working: 19/90 with no subtraction shown scores 0. Pick two multiples whose digits after the point are identical, then subtract.

(b) Work out $2\frac{3}{5} \div 1\frac{1}{4}$. Answer as a mixed number. [2]

1. Improper fractions: $\frac{13}{5} \div \frac{5}{4}$
2. Keep, change, flip: $\frac{13}{5} \times \frac{4}{5}$ **M1**
3. Multiply: $\frac{52}{25}$
4. Mixed number: 52 ÷ 25 = 2 remainder 2

$2\frac{2}{25}$ **A1**

Never divide the whole numbers and the fractions separately (2 ÷ 1 and $\frac{3}{5} \div \frac{1}{4}$). It is always wrong.

Question 3

Negative and fractional indices

(a) Find the value of $\left(\frac{16}{81}\right)^{-3/4}$. [2]

Negative power means flip: $\left(\frac{81}{16}\right)^{3/4}$

Bottom 4 means take the fourth root: $\sqrt[4]{81} = 3$, $\sqrt[4]{16} = 2$, giving $\frac{3}{2}$ **M1**

Top 3 means cube it: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$

$\frac{27}{8}$ (or $3\frac{3}{8}$) **A1**

> **Tip:** always take the root first so the numbers stay small. M1 is also earned for flipping the fraction, or for $\frac{3}{2}$ or $\frac{2}{3}$ seen.

(b) $4^n \times 8 = 2^{11}$. Find the value of n. [2]

Write everything as a power of 2:

$4^n = (2^2)^n = 2^{2n}$ and $8 = 2^3$

$2^{2n} \times 2^3 = 2^{11}$ **M1**

Multiplying means adding the powers: $2n + 3 = 11$

$2n = 8$

$n = 4$ **A1**

> **Check:** $4^4 \times 8 = 256 \times 8 = 2048 = 2^{11}$ ✓

Question 4

Standard form

Work out $(6 \times 10^5) \times (4 \times 10^{-9})$. Give your answer in standard form. [2]

Multiply the front numbers: $6 \times 4 = 24$

Add the powers: $5 + (-9) = -4$

$= 24 \times 10^{-4}$ M1

24 is not between 1 and 10, so fix it:

$24 = 2.4 \times 10^1 \rightarrow 10^1 \times 10^{-4} = 10^{-3}$

2.4×10^{-3} A1

Common mistakes

- Leaving it as 24×10^{-4} earns M1 only.
- Going the wrong way to 2.4×10^{-5}.

Rule to remember: if the front number gets **smaller** (24 → 2.4), the power goes **up** by 1 (−4 → −3). The value stays the same.

Check: $24 \times 10^{-4} = 0.0024 = 2.4 \times 10^{-3}$ ✓

Question 5

Surds

(a) Simplify √75 − √12. [2]

Find the biggest square number that goes into each:

√75 = √(25 × 3) = 5√3 **M1**

√12 = √(4 × 3) = 2√3

5√3 − 2√3

3√3 **A1**

Never write √75 − √12 = √63. You can't subtract under the square root. M1 is for either surd simplified correctly.

(b) Rationalise the denominator and simplify $\frac{4}{3-\sqrt{5}}$. [2]

Multiply top and bottom by the **conjugate** (3 + √5): **M1**

$\frac{4}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$

Top: 4(3 + √5)

Bottom: (3 − √5)(3 + √5) = 9 − 5 = 4

$\frac{4(3+\sqrt{5})}{4}$, then cancel the 4s

3 + √5 **A1**

Why it works: (a − b)(a + b) = $a^2 - b^2$, so the √5 terms cancel and the bottom becomes a whole number.

Question 6

Reverse percentages

A bicycle's price increases by 15%. The new price is $92. Find the original price. [2]

An increase of 15% means the new price is **115%** of the original.

115% = $92

1% = 92 ÷ 115 = $\frac{92}{115} = \frac{4}{5}$ = 0.8 (÷ 23 top and bottom) **M1**

100% = 0.8 × 100

$80 **A1**

Common mistake: taking 15% of $92 and subtracting (92 − 13.80 = $78.20). The 15% is of the **original** price, not the new one, so this scores 0.

M1 is also given for 92 ÷ 1.15 or $\frac{92}{115} \times 100$.

Check: 15% of $80 = $12, and $80 + $12 = $92 ✓

Answers at a glance

Video 1 · Number · 19 marks

Q	Topic	Answers	Marks
1	Prime factors and square numbers	(a) $2^2 \times 3^3 \times 7$ (b) $k = 21$	3
2	Recurring decimals and fractions	(a) $\frac{19}{90}$ (b) $2\frac{2}{25}$	4
3	Negative and fractional indices	(a) $\frac{27}{8}$ (b) $n = 4$	4
4	Standard form	2.4×10^{-3}	2
5	Surds	(a) $3\sqrt{3}$ (b) $3 + \sqrt{5}$	4
6	Reverse percentages	\$80	2
	Total		**19**

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. Next: **Video 2 · Algebra I (Q7–Q11)**.